

Master Device Inventory

Version 1.0 · February 8, 2026

Intent & Workflow

- 1. Capture every load you need to control (HVAC, fans, lights, shades, power, etc.) with product choices, links, and wiring notes.
- 2. Keep this file as the single source of truth; update the relevant section each time we learn something new.
- 3. After each iteration I will call out outstanding questions so you can answer them one by one; once clarified, we update this document.

Categories to Cover

Category	Scope
HVAC	Heat pumps, minisplits, air handlers, and any central heating/cooling units.
All house fans	Whole-house and attic exhaust fans.
Bathroom fans	Exhaust fans serving baths, laundry, and toilets.
Bathroom towel warmers	All electric rails.
Floor heater	Radiant floor or slab heater loads.
Ceiling lights (DALI-2)	DMF downlights and other DALI-native fixtures.
Ceiling fans	Big Ass Fans Haiku series or other architectural fans.
Future pergola heater/fan	Outdoor Haiku pair and pergola heater circuits.
Electric skylights	Motorized skylights + gateways/actuators.
Window shades (TBD)	Motorized shade pre-wire (motors TBD).
SPAN smart panels	Primary and sub-panels for telemetry.
Solar / inverters	Hybrid inverters, rapid shutdown, batteries.
Dumb architectural lights	Pendants, sconces, or any fixtures without embedded DALI drivers.
Seasonal / holiday outlets	Weatherproof outlets reserved for decorative lighting.

Current Device Matrix

HVAC

Device	Qty / Zones	Link	Notes	Source
Carrier D5CURAH60AAK outdoor + D5FUAAH60XAK indoor air-source heat pump	Main floor + bed wing	(spec PDF) docs/SnJ/Heat Pump Spec ID5CURAH60AAK .pdf	60k BTU, inverter, 0–10V compatible; power lands on Sub-SPAN and integrates through LCPs.	docs/electrical_bom_270_bolla.md:521, docs/SnJ/Heat Pump Spec ID5CURAH60AAK .pdf
Fujitsu ASUH12KPAS minisplit + FHMA5X48LOCA fan coil	Tech room / server	docs/SnJ/Spec Mini Split tech room.pdf, docs/SnJ/Fan Coil FHMA5X48LOCA .pdf	Dedicated LCP-2 logic path with Intesis gateway; provides local server cooling.	docs/electrical_bom_270_bolla.md:527, docs/SnJ/Spec Mini Split tech room.pdf

All House Fans

Device	Qty	Link	Notes	Source
QuietCool Stealth Pro 7.0X (ECM motor wide-flow)	2	https://quietcoolsystems.com/whole-house-fan/stealth-pro-x-whole-house-fan/	One per arm of the U-shaped plan; DALI-2 relay/converter logic handled in the LCP but excluded here.	docs/mechanical-ventilation-standard.md:54, docs/electrical_bom_270_bolla.md:177
QuietCool AFG SMT PRO-3.0 gable exhaust fans	4	https://quietcoolsystems.com/attic-fan/smart-gable-attic-fans/	Two per gable end for balanced attic exhaust; interconnected with WHF logic.	docs/mechanical-ventilation-standard.md:62, docs/electrical_bom_270_bolla.md:184

Bathroom Fans

Device	Qty / Rooms	Link	Notes	Source
Fantech FG 6M EC inline fan + LD6 silencer + RSK6 damper	5 dedicated rooms (Kids, Ensuite, Guest, Laundry, Half); primary bath uses shared splitter	https://www.fantech.net/en-us/products/fans-and-accessories/inline-duct-fans/fg/?sku=49900	Uses Southwire NM-B-PCS Duo for 0-10V control; Primary bath has dual diffusers sharing a single motor (per ventilation spec).	docs/electrical_bom_270_bolla.md:189-193, docs/ventilation-spec.md:10-20

Bathroom Towel Warmers

Device	Qty	Link	Notes	Source
Hudson Reed HREH06 (2-towel chrome)	1 (primary bath)	https://www.hudsonreed.com/products/electric-towel-warmers	Large chrome rail; connected to dedicated KNX high-inrush relay.	docs/electrical_bom_270_bolla.md:381
Hudson Reed HREH02 (1-towel chrome)	5 (secondary baths)	same link	Smaller chrome rails for guest/kids baths; grouped into a single relay bank.	docs/electrical_bom_270_bolla.md:382

Floor Heater

Device	Qty	Link	Notes	Source
Schluter DITRA-HEAT electric warming mat & DITRA-HEAT-E-RS1 programmable thermostat	primary bath + water closet plus linear shower entry area	https://www.schluter.com/schluter-us/en_US/Floor-Warming/Thermostats/Schluter®-DITRA-HEAT-E-RS1/p/DITRA_HEAT-E-RS1	Warm floor mat system uses embedded wire under tile; thermostat is Wi-Fi/voice-enabled (Schluter app, Alexa, Google). The thermostat sits on the 240V circuit and handles	docs/ventilation-spec.md, turn1search0, turn1search3

Device	Qty	Link	Notes	Source
	(warm floor not inside shower)		GFCI/floor sensors. Because there is no native KNX interface, automation should instead control a KNX relay/contacter on the 240V feed or ingest the thermostat's data via the Home Assistant bridge—the Schluter unit is not currently KNX compatible.	

Ceiling Lights (DALI-2)

Device	Qty	Link	Notes	Source
DMF Lighting X-Series Square Flangeless downlight assembly	80 (core recessed)	https://www.dmflighting.com/product/x-series/	Includes housing (X2NCS), mud-in plate (X2KSMUD), trim (X2TSDSWHFL), and LED module (XMD12930WFD); each is DALI-2 addressable.	docs/electrical_bom_270_bolla.md:211, catalog.json:650-839
DMF X-Series Square Flangeless Wet	4 (shower areas)	same link	Wet-rated trim variant (X2TSDSWHFL-WET).	docs/electrical_bom_270_bolla.md:212, catalog.json:891-965

Ceiling Fans

Device	Qty	Link	Notes	Source
Big Ass Fans Haiku 52" aluminum (fan + light driven)	3	https://www.bigassfans.com/fans/haiku/	Project standard for ceiling fans; each uses 0-10V fan and light channels via DALI-2 converters (not listed here).	docs/electrical_bom_270_bolla.md:213, snapshot_bom.json:62, catalog.json:59

Future Pergola Heaters & Fans

Device	Qty	Link	Notes	Source
Pergola heating circuits (3 × 6 kW each)	3	https://bromic.com/us	Candidate: Bromic Tungsten Smart-Heat 6000W/30A (or equivalent high-power outdoor infrared heater); wiring already splits into 4 zones, so plan to control each heater via KNX relay outputs (or	docs/electrical_bom_270_bolla.md:332, docs/electrical_automation_master_plan.md:111

Device	Qty	Link	Notes	Source
			a KNX-to-RF gateway) with the custom RF hub providing feedback/status.	
Outdoor Haiku fans (pergola)	2	https://www.bigassfans.com/fans/haiku/	Two fans pre-wired for client selection (presumed Big Ass Fans Haiku); integrate the RF fan control into KNX by mapping per-zone relays so cloud/KNX scenes can mirror the custom RF hub.	docs/electrical_bom_270_bolla.md:333

Electric Skylights

Device	Qty	Link	Notes	Source
Velux VSE M08 (~2×4)	1	https://www.velux.com/products/roof-windows/solutions/smart-home	Hallway skylight (Group 1) – high-quality flush-mounted unit intended for standing-metal roof; integrating via Velux Integra gateway (KNX-compatible) for easier automation.	docs/knx-dali-standard.md:313-319
Velux VSE C01 (~2×2)	7	same family link	Dining (6) + Laundry (1) skylights; flush mount and call out Velux Integra (KNX-ready) controller to keep integration tight with KNX/automation.	docs/knx-dali-standard.md:314-320

Window Shades (TBD)

Device	Qty	Link	Notes	Source
Motorized window shades (pre-wire only)	home-run cables to every window header	can't find / TBD	Pre-wire uses 16/4 shielded cable (LV power+feedback) to allow Lutron/Somfy/Hunter Douglas/etc.; keep each run terminated in labeled "shade vaults" at LCPs. No line-voltage runs needed now, so future controller choices remain flexible.	docs/electrical_bom_270_bolla.md:295, docs/device-system-data-models.md:802-818

SPAN Smart Panels

Device	Qty	Link	Notes	Source
SPAN Gen 2 Smart Panel (32-circuit)	2	https://span.io/products/main-32	Main service panels with Ethernet telemetry; routed to automation enclosures.	docs/electrical_bom_270_bolla.md:35, docs/electrical_automation_master_plan.md:55
SPAN Sub-panel (16-circuit)	1	https://span.io/products/panel/16	Dedicated to logic/tech room loads; part of Phase 1.	docs/electrical_bom_270_bolla.md:36

Solar / Inverters / Energy Wall

Device	Qty	Link	Notes	Source
Hybrid inverter (EG4 18kPV / Sol-Ark 15K preferred)	2 planned (paralleled)	https://www.eg4.com/ or https://solarkpowersystems.com	Phase 2/3 Energy Wall will house paralleled inverters with battery management; slots reserved behind SPAN panels.	docs/electrical_automation_master_plan.md:28-64, catalog.json:757-774, snapshot_bom.json:38
Battery bank (EG4 PowerPro wall mount)	1 bank	https://www.eg4.com/	Placed below inverter zone; size to match inverter rating.	docs/electrical_automation_master_plan.md:64, catalog.json:798-800, snapshot_bom.json:30

Dumb Architectural Lights

Device	Qty	Link	Notes	Source
Decorative pendants (non-DALI), phase-dimmable 120V	9 placeholder locations	https://integratorpro.app/placeholders	Standardize on phase-dimmable LED bulbs; install junctions sized for DALI pucks if future lighting upgrades are desired.	docs/electrical_bom_270_bolla.md:222, catalog.json:723, snapshot_bom.json:118
Wall sconces (non-DALI), phase-dimmable 120V	12 placeholder locations	same placeholder link	Use 120V dimmable lamps now; allow future DALI integration by keeping box depth/depth and wiring cleared.	docs/electrical_bom_270_bolla.md:233, catalog.json:733, snapshot_bom.json:126

Seasonal / Holiday Outlets

Device	Qty	Link	Notes	Source
Weatherproof GFCI outlets (eave)	2	can't find / TBD	Outdoor-rated GFCI duplex feeds, tied to KNX high-inrush relay for automated schedule control.	docs/electrical_bom_270_bolla.md:346
Weatherproof GFCI outlets (ground)	2	can't find / TBD	Ground-level weatherproof GFCI duplex with KNX relay control; four total outlets service holiday lighting.	docs/electrical_bom_270_bolla.md:347

Outstanding Questions

2/8/26, 7:38 AM

None—everything in this doc now has a proposal/action. Keep me posted if you need further refinements.